Beyond group averages: individual differences in infant auditory habituation timing using high-density diffuse optical tomography in rural Gambia.

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Abstract

Habituation, or reduced neural response to repeated stimuli, is an early-life marker of learning and development. Prior infant functional near-infrared spectroscopy studies investigating this phenomenon relied on group-averaged data that obscure individual differences. This study used high-density diffuse optical tomography, which offers denser spatial sampling, to enable individual-level habituation analysis in 250 Gambian infants (INDiGO Trial) at 6 and 12 months. Infants completed an auditory habituation-and-novelty-detection paradigm; data were analysed at the individual level, using threshold-free cluster enhancement for spatial and temporal significance, followed by functional change point detection for timing of response change. Only 13-20% of infants showed a significant individual-level evoked response but, among these, 58-83% showed habituation in spatially inconsistent locations across infants, with the masking at the group level therefore likely due to the small number of infants exhibiting an evoked response to the stimulus initially. Functional change point detection showed that 12-month-olds habituated ~3 trials earlier than 6-month-olds, replicating and extending earlier group-level findings, with the effect driven primarily by the HbO signal. Findings demonstrate the value of individual-level high-density diffuse optical tomography, threshold-free cluster enhancement, and functional change point detection, for uncovering developmental habituation markers otherwise masked by group averaging, while highlighting the need for further work to understand the low 'responder' rate.

Keywords functional near-infrared spectroscopy, high-density diffuse optical tomography, infancy, habituation, neurodevelopment, longitudinal, functional change point detection

Highlights

- Only 13–20% of infants showed significant individual-level haemodynamic responses ('responders')
- Among responders, 58–83% showed clear habituation effects ('habituators')
- 12-month-olds habituated to an auditory stimulus significantly earlier than 6-month-olds
- Habituation effects were spatially inconsistent, explaining prior null group-level results
- Age effect was chromophore-specific, driven mainly by the oxyhaemoglobin signal

1 Introduction

1.1 Habituation and its developmental importance

Habituation, characterised by a diminishing response to a repeated stimulus is an early-developing process of learning related to the building of internal representations of repeated stimuli [1], [2], [3], [4] and linked to cognitive resource conservation and early survival [5]. The closely-related concept of dishabituation, or novelty-response, is an increased response to stimuli which do not fit these internal representations, thus enabling the prioritisation of novel and potentially hazardous [3]. The two phenomena are closely related and thought to direct attention in support of early learning [3], [5].

The number of stimulus repetitions required for an infant to fully habituate negatively correlates to cognitive functioning in later developmental stages [6], [7]; as repeated stimuli are typically presented at a fixed rate within a habituation paradigm, a lower number of repetitions required to habituate is functionally equivalent to an earlier onset of habituation, motivating a focus on habituation timing as a marker of developmental change. Habituation has consequently been found to be exhibited in older infants, at later stages of early development, to a greater degree than younger counterparts in both cross-sectional and longitudinal studies, in both humans and non-human mammals [8], [9], [10]. Furthermore, habituation does not appear to be universal to all developing infants, meaning it may be appropriate to report analyses into data collected from a carefully selected subset of infants rather than the entire collected sample [11], [12] – a point returned to below in relation to individual-level analysis.

1.2 Optical neuroimaging for infant habituation research

Habituation is a natural fit for infant neuroimaging research because it does not rely on overt behaviours, and its outcome of interest is cognitive rather than behavioural [5]; see Nordt and colleagues [13] for a detailed discussion. Indeed, past infant functional near-infrared spectroscopy (fNIRS) research shows that the evoked infant cortical hemodynamic response can gradually decrease when stimuli are repeated [3], [14], [15], [16].

fNIRS and high-density diffuse optical tomography (HD-DOT) are two closely-related neuroimaging modalities. Both use near-infrared light changes between light sources and detectors to measure concentration changes of haemoglobin, when bound to oxygen molecules (oxygenated: HbO) and in the unbound form (deoxygenated: HbR) [17], [18], [19]. fNIRS headgear arrays typically contain fewer than 100 channels, with each one designed to sample data from a different cortical location: ‘sparse array’ imaging, suitable for topographical analysis [20]. HD-DOT is an extension of fNIRS which employs imaging arrays of a greater density of channels with differing lengths, enabling tomographic reconstruction of cortical haemodynamic changes. Both are well-suited to infant studies [21], [22] and can be used to investigate infant brain function, providing objective measures and overcoming limitations associated with behavioural assessments and alternative neuroimaging modality approaches [23], [24].

However, near-infrared imaging data is frequently masked by physiological noise and motion artefacts [25], [26], [27], [28]; analyses for sparse array data are often therefore conducted on data after cross-participant averaging of the signal, which reduces the influence of these unwanted components [29]. This precludes analysis of individual infant NIR imaging data collected during a key developmental window and impedes the identification of early neurodevelopmental markers for informing targeted early-life interventions. In contrast, the spatial coverage of HD-DOT permits sampling of similar cortical locations by multiple channels, thereby reducing the need for cross-participant averaging [30], [31] and increasing the likelihood of meaningful individual-level analyses.

1.3 Towards individual-level habituation analysis

Habituation of haemodynamic responses has previously been identified in Gambian infants during the first year of life using an auditory habituation and novelty detection ('HaND') task administered by the BRIGHT study, both in cross-sectional and longitudinal analyses [5], [9], [10], [16], [32]. These works used block averaging [29], threshold-free cluster enhancement (TFCE) [9], [32], [33], and functional change point (FCPt) detection [34], [35] methods to determine channels where significant responses occurred and changed throughout HaND paradigm [16]. The observed habituation of Gambian infants in the first year of life persisted, even when introducing a novel stimulus intended to elicit a heightened response, indicating continued habituation. However, these findings were based on cross-participant-averaged haemodynamic responses collected using a sparse fNIRS array which limited the insights to the group-level, potentially biasing results by infants who did not respond to the paradigm as well as precluding insight into individual-level habituation.

TFCE provides a means of identifying spatially extended regions of significant haemodynamic activation while accounting for the temporal and spatial structure of functional neuroimaging data [32], [33]. However, whilst well-suited to identifying where significant responses occur, TFCE does not directly characterise when these responses change across repeated stimulus presentations.

FCPt detection provides a means of characterising the timing of significant changes in the haemodynamic response across repeated stimulus presentations, and is therefore well-suited to investigating the timing of habituation [10], [34], [36], [37]. However, whether the developmental effects identified using FCPt detection in group-averaged sparse-array data can be replicated at the individual level remains an open question, and one that may be answered using HD-DOT.

The Improving infant Neurocognitive Development and Growth Outcomes with micronutrients ('INDiGO') Trial [38], which followed the BRIGHT study in the same testing location in rural Gambia, utilises HD-DOT imaging to assess infants during a similar HaND paradigm. Preliminary analyses on these data using cross-infant-averaged haemodynamic responses identified significant task-evoked responses to the HaND paradigm in auditory-associative regions in 16 and 43 parcel × chromophore combinations at 6- and 12-months of age (hereafter '6mo' and '12mo') respectively, yet found evidence for significant habituation in only 2 combinations at 12mo and none at 6mo, with no significant novelty response

detected at either age [zxcv, methods paper; see also Supplementary Materials S1].

While this may point to a lack of habituation in this separate cohort, the wider literature suggest that there may be a subset of infants who exhibit habituation ('habituators') whose responses are being obscured when combined with those of the entire sample of infants [11], [12]. Such heterogeneity may reflect individual differences in neural maturation, arousal state at testing, or early-life exposures known to shape sensory processing more broadly [39], [40], [41], [42], any of which could produce a subgroup whose signal is diluted rather than absent at the group level. The HD-DOT dataset collected as part of INDiGO, and the limitations of the aforementioned previous group-averaged findings in both BRIGHT and INDiGO, provide the opportunity and motivation, respectively, to investigate habituation at the individual level. This work does so using both TFCE and FCPt detection with the aim of examining how these processes vary across infants during a key period of neurocognitive development.

1.4 Objectives

This work investigates the following two research questions:

(a) Is there evidence for individual differences in the presence and change of evoked haemodynamic responses to a repeated speech paradigm in Gambian infants at 6- and 12mo of age?

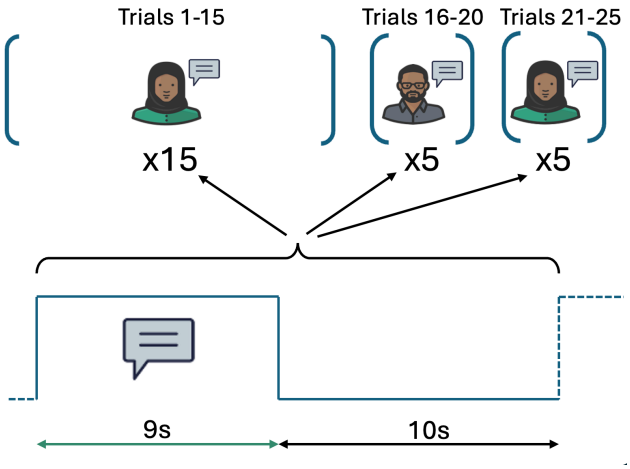
(b) Does the timing of habituation differ between individual Gambian infants at 6- and 12mo, and is habituation timing associated with age?

It was hypothesised that individual differences in the presence of evoked haemodynamic responses and habituation would be evident, and that infants at 12mo would exhibit earlier habituation than infants at 6mo.

2 Methods

Participants were 239 infants recruited from rural Gambia as part of the INDiGO Trial [38], assessed at 6- and 12mo of age. During each visit, infants were presented with an auditory HaND paradigm comprising 25 trials of a short spoken stimulus. The same recording with a female voice was used for trials 1-15 – the Habituation trials, split during analyses into Fam1 (trials 1-5), Fam2 (trials 6-10), and Fam3 (trials 11-15). Following this a male voice was used for trials 16-20 (the Novel trials); finally, the original recording from trials 1-15 was again repeated for trials 21-25 (the Post-Test trials). HD-DOT data were acquired using the LUMO wearable system (Gowerlabs Ltd., UK) [21], [43] at 735- and 850nm. The imaging array covered bilateral temporal and frontal cortical regions. Full details, plus description of the preprocessing, image reconstruction, and aggregation of reconstructed time series into cortical parcels are described in [methods paper, zxcv]. Here we describe in detail only the statistical analyses performed on per-infant and per-[infant × parcel] time series. Unless otherwise stated, statistical analysis was implemented in Python 3.10.14 [30] using in-house, publicly-available code.

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Cap on infant photo to go here when Bea gets back from leave!

Figure 1. Top: schematic of the HaND paradigm, with 25 trials consisting of spoken sentences (8s) and inter-stimulus intervals of 10s. The speaker is changed for trials 16-20 to induce novelty-responses. Bottom: Description of array coverage schematic and/or infant photos with headgear when copied in.

2.1 Threshold-free cluster enhancement (TFCE)

Individual-level TFCE implementation

To test whether individual infants showed detectable evoked responses at the parcel level, TFCE [9], [32], [33] was applied with spatiotemporal integration and empirical null distributions constructed via permutation.

For each infant × chromophore × age combination, a [parcels × time points]-dimension t-statistic map was constructed; each entry t_{ij} in the *t*-map held the *t*-statistic for parcel i at time point j. Significant haemodynamic responses were assessed by comparing per-trial amplitude to baseline using data from Fam1. For condition contrasts, each t_{ij} held the between-epochs *t*-statistic comparing two 5-trial epoch groups: Fam1 vs Fam3 for standard habituation; Fam1 vs Novel for

delayed habituation; and Fam1 vs Post for 'long-range' habituation. For completeness, testing for novelty-response was also conducted, by comparing the Fam3 vs Novel epochs. Infants were required to have ≥ 3 valid trials for Fam1 (significant responses) or in each condition epoch (contrast tests) being compared; any infant × chromophore × parcel × age combination (hereafter 'observation') with an insufficient number of trials was excluded from that contrast's TFCE test. TFCE was then applied to this observed *t*-map to enhance spatially and temporally extended clusters.

The TFCE algorithm iterates over a range of candidate thresholds, *h*. For entry t_{ij}, the TFCE value is computed using the integral $TFCE(t_{ij}) = \int_{h=0}^{t_{ij}} e(h)^E h^H dh$, where *e(h)* is the cluster extent (the number of neighbouring entries connected to the entry of interest that also survive at threshold *h)*; and E and *H* are extent and height exponents [33]. Entries embedded in extended clusters therefore accumulate large TFCE values across many thresholds; isolated peaks receive comparatively small values. Entries were considered neighbours if they corresponded to adjacent parcels at the same time point (spatial neighbours from the parcel adjacency graph), the same parcel at adjacent time points (temporal neighbours), or spatially adjacent parcels at adjacent time points (diagonal neighbours) [44]. Integration was performed at 50 threshold steps from zero to the maximum observed statistic; extent and height exponents were set at *E*=1 and *H*=2 [9], [45].

Empirical null distributions for TFCE

Statistical inference was based on empirical null distributions constructed separately for activation and contrast tests.

To determine significant responses, each trial's time series in Fam1 was independently circular-shifted by a random number of samples, preserving within-trial temporal structure while disrupting the alignment between trial onset and evoked response. The full pipeline (*t*-statistic map + spatiotemporal TFCE) was re-run on each permuted dataset *N*=1000 times, yielding *N* null TFCE maps.

For subsequent condition contrast tests, trial labels (which group each trial belonged to) were exhaustively enumerated across all possible relabellings of the two 5-trial epochs, therefore preserving trial-level temporal structure while disrupting the group assignment. The full pipeline was re-run on each relabelled dataset, yielding a null TFCE distribution comprising all valid permutations for that infant × chromophore × parcel × age.

For each infant × chromophore × parcel × age combination, the *p*-value was computed as (number of null |TFCE| values ≥ |observed TFCE| + 1) / (N + 1); *p*-values were then Benjamini–Hochberg FDR-corrected [46], [47] within each infant × chromophore × parcel × age map at $q = 0.05$.

Directional filtering and prerequisite chain testing

The individual-level TFCE analysis was two-sided. Significance was then filtered post-hoc to retain only results in the hypothesised direction for the given chromophore. For significant responses, this meant a positive *t* for HbO and negative *t* for HbR. For condition contrasts, positive *t* indicates a larger response in the first-listed epoch (e.g. Fam1 > Fam3 for canonical habituation in HbO data).

Prerequisite chain testing enforced biological plausibility: contrast tests were considered for an infant × chromophore × parcel × age only if activation was also significant at that infant × chromophore × parcel × age, and the temporal windows of significance for the contrast test overlapped with those of the significant response. This ensured only those infants with significant initial responses ('responders') could subsequently be classed as 'habituators'. For the novelty contrast (Fam3 vs Novel), a further prerequisite required habituation-masked significance at the same infant × chromophore × parcel × age.

2.2 Functional change point (FCPt) detection

Implementation of FCPt detection

FCPt detection was conducted on individual-level data exhibiting evidence for habituation of the haemodynamic response compared to the first 5 (Fam1) trials, identified using TFCE. Namely, infant × chromophore × parcel × age combinations with significant standard- (Fam1 > Fam3), delayed- (Fam1 > Novel), or long-range (Fam1 > Post) habituation contrasts were retained for further analysis. FCPt detection was applied to the 25-trial sequence, treated as a functional time series, using the fully-functional detection approach developed by Aue and colleagues [10], [36] with change points detected in the mean function; full details of the FCPt implementation are in Appendix A. Infant × chromophore × parcel × age combinations missing at most 4 non-consecutive trials were assessed. Each change point was characterised as decreasing or increasing based on whether the response increased or decreased following the FCPt. This was determined by calculating the mean haemodynamic response before and after each statistically significant FCPt, then calculating the definite integral of the curve's absolute values using Simpson's rule, via *scipy.integrate.simpson* [48].

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The following FCPt groups were extracted from those identified across all tested infant × chromophore × parcel combinations at each age:

(i) Decreasing change points
(ii) Decreasing change points which were also the first to occur in a channel ('First And Decreasing')
(iii) Decreasing change points which were also the last to occur in a channel ('Final And Decreasing')
(iv) The earliest occurring decreasing change points in a channel ('Earliest Decreasing')
(v) The latest occurring decreasing change points in a channel ('Latest Decreasing').

Statistical tests on change points

Statistical testing was conducted on the different FCPt sets to examine age-related effects.

Following the analytical framework used in prior sparse-array group-averaged fcpt work in the same population [10], ordinal logistic regression (OLR) with Age as the fixed effect was applied to each retained change-point group as the primary test. The model used the logit link function, implemented via OrderedModel from *statsmodels* [46] and optimised via the Newton method.

Following the OLR fit, a Wald test on the Age coefficient confirmed significant age effects [49], and Tukey Honest Significant Difference (HSD) tests were applied to obtain the between-age mean difference in change-point trial number [50].

OLR *p*-values were corrected for multiple comparisons using the Holm–Bonferroni method [51] across the five tests, applied separately within each chromophore. Wald and Tukey HSD statistics were reported uncorrected as follow-ups.

Robustness analyses: widening the testing pool

The primary FCPt detection was performed on habituation in at least one of the Fam1 vs Fam3, Fam1 vs Novel, or Fam1 vs Post contrasts. (defined by individual-level activation TFCE significance). To test the robustness of the FCPt findings to subset definition, all analyses were repeated on the subset of infants who exhibited a significant haemodynamic response, regardless of subsequent habituation condition contrast results. Statistical tests on change-point trial numbers were repeated for this wider subset, and results compared to the activation-only primary analysis.

Sensitivity analyses: chromophore-specific testing

To understand whether chromophore-specific FCPt findings were driving any significant cross-chromophore results, all analyses were repeated on both HbO and HbR data. Statistical tests on change-point trial numbers were repeated separately for HbO and HbR data across infant × parcel combinations at each age, and results compared to the cross-chromophore primary analysis.

3 Results

3.1 Attrition and per-test sample sizes

Of the starting sample of 250 infants, 11 were withdrawn from the study by the time of the 6mo imaging session. Of these, 200 (83.6%) and 178 (74.4%) were retained in the final 6- and 12mo analyses respectively, following further exclusions for testing non-completion, technological issues, and data quality (Figure 2).

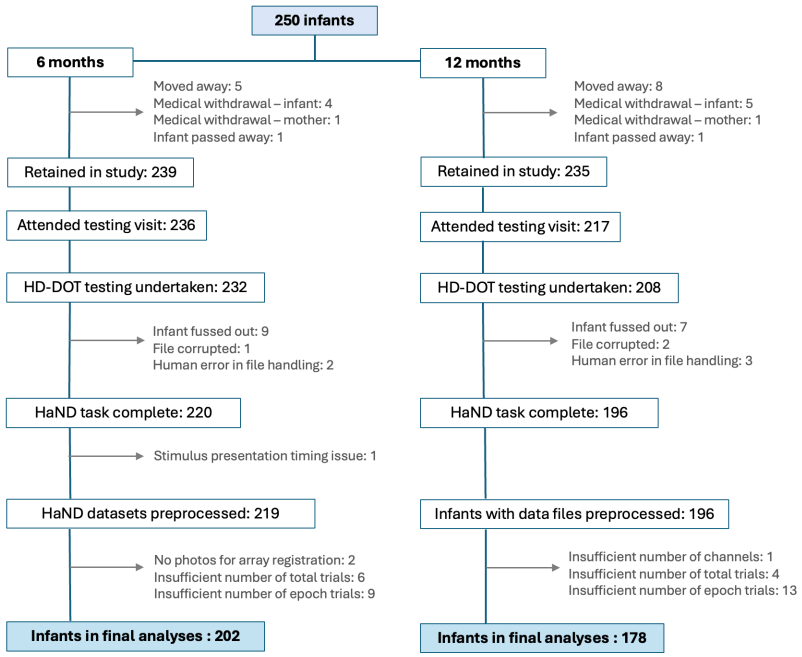


Figure 2: data retention and attrition across the two timepoints, plus reasons for data loss at various stages of data collection, management and preprocessing.

Infants reached the individual-level, per-parcel statistical analysis stage if three trial time series were available for an age × chromophore combination in the first epoch (trials 1-5), enabling the test for a significant haemodynamic response. Sample sizes at the median parcel were 195-196 infants at 6mo and 174 infants at 12mo for this test.

Each subsequent condition contrast test was at least conditional on haemodynamic response significance (in the case of habituation, delayed habituation, and long-range habituation) and required a minimum of three trial pairs for the relevant contrast; the novelty contrast was additionally conditional on habituation significance. Consequently, per-parcel sample sizes for these contrast tests were substantially smaller: median *n* per age × parcel × chromophore combination dropped to 6-8 infants for habituation, delayed and long-range habituation, and to 7–8 for novelty response (Table 1).

Table 1. Individual-level TFCE results: whole-brain summary

		HbO			HbR		
	Age (mo)	**Tested**	**Sig.**	**%**	**Tested**	**Sig.**	**%**
Significant response (Fam1)	6	202	30	14.9	202	27	13.4
	12	178	36	20.2	178	35	19.7
Habituation (Fam1 vs Fam3)	6	23	19	82.6	19	15	78.9
	12	26	20	76.9	24	14	58.3
Delayed Hab (Fam1 vs Novel)	6	26	15	57.7	19	15	78.9
	12	26	20	76.9	26	17	65.4
Long-range Hab (Fam1 vs Post)	6	20	14	70.0	17	16	94.1
	12	21	14	66.7	21	17	81.0
Novelty (Fam3 vs Novel)	6	26	7	26.9	22	5	22.7
	12	26	5	19.2	25	6	24.0

For activation, 'Tested' is the number of unique infants tested (summed across parcels where each infant had testable data). For condition contrasts, 'Tested' is the number of infants who cleared the activation prerequisite. 'Sig.' is the number reaching individual-level TFCE significance (FDR-corrected, q < 0.05) with the appropriate directional filter.

3.2 Individual-level TFCE analyses

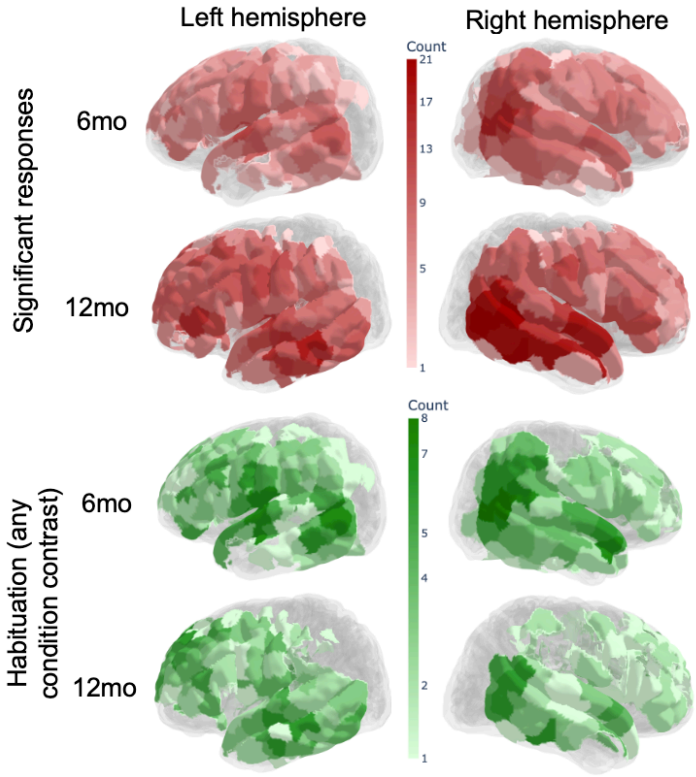
Significant responses

A significant evoked response was detected in a modest proportion of infants at both ages. Summed across parcels (i.e. each infant contributing wherever they had testable data), 14.9% of tested infants at 6mo showed HbO activation and 13.4% HbR; at 12mo, 20.2% HbO and 19.7% HbR (Figure 3).

Full methods and results for the group-level analysis are reported in the companion methods paper [methods paper, zxcv] and summarised in Supplementary Materials S1. The concordance between these previously obtained group-level results and individual-level significance results obtained here using TFCE was examined formally. Mann–Whitney U tests comparing four individual-level metrics – the (i) number and (ii) proportion of individually-significant infants, and median individual (iii) *t*-statistic and (iv) TFCE peak – showed that group-significant parcels scored consistently higher on all four metrics than group-non-significant parcels across all four age × chromophore combinations (e.g. median 13–17 vs 4–6 individually-significant infants; all Holm-Bonferroni-corrected $p < 0.001$), with no corresponding difference in the number of infants tested per parcel at three of four combinations. This indicates that group-level tests succeed when individual-level effects converge spatially, and fail when they do not, independent

of sample size (Supplementary Materials S2 for details, including additional logistic regression).

Figure 3. Number of infants with significant haemodynamic responses in the Fam1



trials (top, red) and habituation (bottom, green) for HbO data. Significant evoked responses and habituation are spatially varied across the coverage of the imaging array at both ages, with higher clustering in the temporal lobe and superior temporal sulcus of the right hemisphere. Analogous HbR images can be found in Supplementary Materials S3.

Habituation

Among infants who cleared the activation prerequisite, habituation (Fam1 > Fam3) was detected in 58–83% of eligible infants at 6- and 12mo (HbO: 82.6% at 6mo, 76.9% at 12mo; HbR: 78.9% at 6mo, 58.3% at 12mo). Detection rates for delayed- (Fam1 > Novel) and long-range (Fam1 > Post-test) habituation among eligible infants were comparable to standard habituation: 57.7–78.9% for delayed habituation and 40–94% for long-range habituation, with the highest rate (94.1%) in HbR at 6mo for the long-range condition contrast. Despite these high individual-level detection rates, concordance testing confirmed that individual habituation

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effects did not converge spatially across infants: no habituation contrast was associated with significant parcels at the group level for any age × chromophore combination.

Given this spatial heterogeneity of individual habituation effects, per-infant response parcel sets were constructed to enable visualisation and assessment of the haemodynamic response independent of fixed anatomical locations. For each infant × chromophore × age combination with at least one activation-significant parcel, the response parcel set was defined as the union of that infant's significant parcels, and a single aggregate waveform was computed by averaging across these parcels. Figure 4 presents the group-averaged aggregate waveforms (± SEM across infants) for Fam1 and each contrast epoch at both ages, confirming canonical habituation morphology: a clear HbO/HbR increase/decrease, respectively, during Fam1 with reduction in response size across Fam3, Novel, and Post-test epochs at both 6- and 12mo.

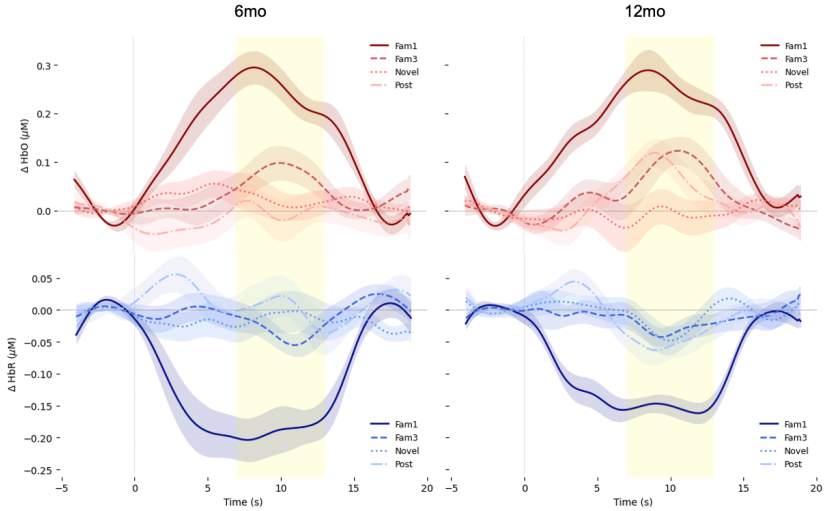


Figure 4: Mean Fam1, Fam3, Novel and Post-Test Habituation waveforms for 6mo (left) and 12mo (right) infants. Waveforms are composed by taking the mean of per-infant response parcel set waveforms, for infant × parcel × age × chromophore combinations with significant responses. Clear evoked haemodynamic response morphology is exhibited for Fam1 data, with subsequent epochs exhibiting either smaller concentration changes or no change at all.

Novelty-response

Among infants who cleared both prerequisites (significant response in Fam1, and Fam1 vs Fam3 habituation), a significant novelty response was detected in between 6-8 eligible infants across all age × chromophore combinations. This amounts to 19–27% of eligible infants: a lower conditional rate than any of the habituation contrasts.

3.3 Functional change point detection

Cross chromophore findings

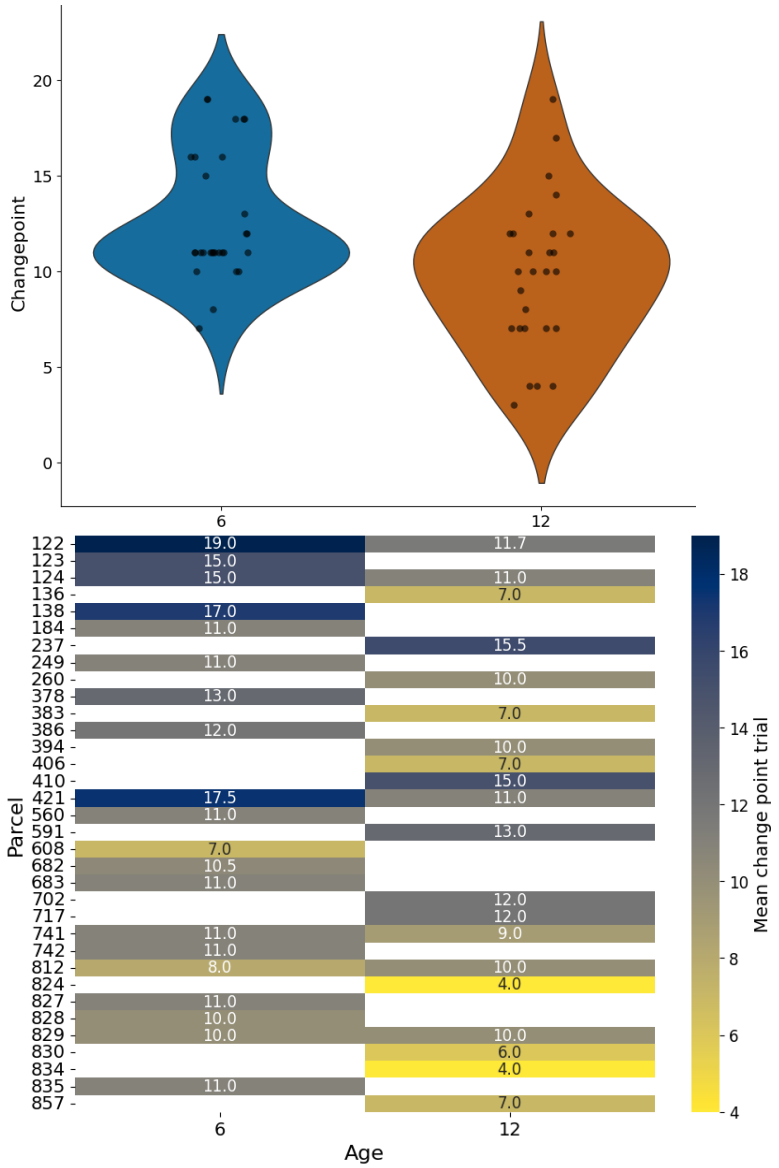
Applied to infant × chromophore × parcel × age combinations in infants who showed TFCE-significant habituation in at least one contrast (n = 56), Final And Decreasing change points occurred significantly earlier at 12mo than at 6mo (OLR: β = -0.216, *p(corrected)* = 0.048; Tukey HSD mean difference = -2.87 trials, Tukey p = 0.005; see Figure 5). Age had no significant effect on any other FCPt category after correction (Table 2), though the Latest Decreasing category also exhibited directionally consistent age effects which did not survive correction (n = 61; OLR: β = -0.150, *p(corrected)* = 0.055*;* Tukey HSD mean difference = -2.10 trials, p = 0.039).

Table 2. Ordinal logistic regression and follow-up tests on FCPt trial number by age

		Ordinal logistic regression			Wald test	Tukey HSD (12 vs 6mo)	
	FCPt category	**β**	**p (raw)**	**p (Holm)**	**p**	**Δ trials**	**p**
Cross-chromophore	Decreasing	-0.1112	0.1225	0.3674	0.1225	-1.5139	0.1182
	First and decreasing	-0.1083	0.2012	0.3922	0.2012	-1.4836	0.1441
	Final and decreasing	**-0.2158**	**0.0095****	**0.0475***	**0.0095****	**-2.8723**	**0.0048****
	Earliest decreasing	-0.0994	0.1961	0.3922	0.1961	-1.1275	0.2493
	Latest decreasing	-0.1499	0.0545	0.2181	0.0545	-2.0893	0.0392
HbO	Decreasing	-0.2309	**0.0184***	0.0735	0.0184	-2.8125	0.0129
	First and decreasing	-0.2088	0.0549	0.0867	0.0549	-2.2274	0.0408
	Final and decreasing	**-0.3207**	**0.0051****	**0.0256***	**0.0051****	**-3.8632**	**0.0017****
	Earliest decreasing	-0.2255	**0.0289***	0.0867	0.0289	-2.2417	0.0256
	Latest decreasing	-0.2223	**0.0302***	0.0867	0.0302	-2.8500	0.0196
HbR	Decreasing	-0.0219	0.8508	1	0.8508	0.0000	1
	First and decreasing	-0.0151	0.9179	1	0.9179	-0.4861	0.8262
	Final and decreasing	-0.1465	0.2651	1	0.2651	-1.9333	0.2536
	Earliest decreasing	-0.0354	0.7812	1	0.7812	-0.2167	0.9134
	Latest decreasing	-0.1465	0.2651	1	0.2651	-1.9333	0.2536

*OLR with Age as fixed effect applied to each FCPt category. OLR p-values Holm–Bonferroni corrected across 5 tests within each chromophore group. Wald and Tukey HSD reported uncorrected as follow-ups. β = OLR coefficient for Age; negative values indicate earlier change point trial numbers at 12 months compared to 6 months. Δ trials = Tukey HSD mean difference (12mo − 6mo); negative values give approximate trial number differences between ages for each FCPt category. * signifies p < 0.05; ** signifies p< 0.01.*

This age effect on 'final and decreasing' FCPts was replicated during the robustness testing, when analysis was widened to infants who exhibited a significant haemodynamic response, regardless of whether significant habituation of this response was uncovered when conducting further condition contrasts (n = 57; OLR: β = -0.224, *p(corrected)* = 0.034*;* Tukey HSD mean difference = -2.95 trials, p = 0.003). As with the primary results, age also had no significant effect on any other FCPt category (Supplementary Materials S4).



459 *Figure 5. Top: violin plot containing the distribution of final and decreasing change*
460 *points at 6mo vs 12mo. Bottom: heatmap containing the mean HaND paradigm*
461 *trial number per parcel of the final and decreasing FCPts at 6- and 12mo.*

462

Prior individual-level TFCE analyses at the parcel level showed a small number of infants at any chromophore × age reaching significance for the novelty-response (Fam3 vs Novel) contrast. Novelty-related FCPt detection analyses were therefore undertaken by assessing the effect of age on increasing FCPts; however, detected FCPts produced inconclusive results for Age effect, possibly due to small sample sizes.

Chromophore-specific findings

For both chromophores, the statistical analyses were repeated for infant × parcel × age combinations in infants who showed TFCE-significant habituation in at least one contrast. Final And Decreasing HbO change points occurred significantly earlier at 12mo than at 6mo (n = 34; OLR: β = -0.3207, $p(corrected)$ = 0.03; Tukey HSD mean difference = -3.86 trials, Tukey p = 0.002). Every other FCPt category exhibited non-significant age effects for HbO after correction for multiple comparisons, however 3 of the remaining 4 categories ('Decreasing', 'Earliest Decreasing', and 'Latest Decreasing') showed nominally significant OLR and Tukey HSD age effects in the same direction, before correction (Table 2). All FCPt categories for HbR data were non-significant prior to correction.

4 Discussion

4.1 Research question (a): responders and habituators

The small sample sizes for investigations into habituation – using both TFCE and FCPt detection – are striking, particularly given the initial sample size of infants (*n* = 250) used in this study. Data from at least 71% of infants underwent TFCE testing for significant haemodynamic responses at both ages (Table 1), indicating that infant- and signal-driven data attrition prior to statistical analyses were within a typical range for infant fNIRS studies [52]. Interestingly, the stage of TFCE analyses during which the largest proportion of infants was excluded from subsequent analyses was after the initial testing for significant haemodynamic responses in trials 1-5 (Fam1), with a maximum of 20.2% of infants – 36 of 178 for HbO data at 12mo – exhibiting a significant response. In contrast, of those infants exhibiting a significant response in the initial trials, a higher proportion subsequently exhibited significant habituation, with a minimum of 57.7% significance for all age × chromophore combinations across all three habituation condition contrasts.

This supports the previous literature which posits the existence of a subsection of 'habituators', whose responses are obscured when combined with those of the entire sample of infants [11], [12]. More importantly, given the associated attrition rate, it may also provide evidence for a much smaller proportion of 'responders'. In studies with a smaller sample size, such small proportions of infants responding to the task paradigm may have real and unwanted experimental consequences which may lead to false negatives, particularly when conducting analyses at the group-level. An alternative explanation is equally relevant for the wider infant fNIRS/HD-DOT community: it is possible that the small proportion of apparent 'responders' is signal-driven. The introduction of HD-DOT hardware has also introduced a greater number of necessary methodological choices – such as image reconstruction parameters, cortical parcellation choices, and cortical registration approaches – all of which may negatively impact downstream analyses [53], [54], [55], [56]. If HD-DOT is to fulfil its potential as a tool for developmental research [57] it is vital that processing and statistical methodology for infant data keeps pace with the advances in hardware, particularly given the challenges in applying approaches designed with adult participants in mind to infant participants [27], [58].

4.2 Novelty response in the first year of life

Individual-level TFCE results for all four age × chromophore combinations here also found evidence for a number of individuals who exhibited a novelty response (Table 1), though the small sample size precluded further analyses using FCPt detection. Nevertheless, these findings contradict previous group-level work in a sample from the same population of similar ages, which found no evidence for novelty-response in infants of similar ages [9]. The small number of infants responding in the expected manner to the novel stimulus would explain why this was not detected in cross-infant, group-level haemodynamic responses. This points to a further advantage of using individual-level approaches for analyses

using HD-DOT imaging hardware, and motivates a possible future avenue for research into this phenomenon in younger infants.

4.3 Research question (b): age differences in individual auditory habituation timing

This work represents the first individual-level evidence for developmental differences in infant auditory habituation timing using HD-DOT. Findings show that 12mo infants completed adaptation to the auditory stimulus trials ~ 3 trials earlier than at 6mo, corresponding to a Tukey HSD age effect of -2.87, alongside evidence for age-independence in habituation onset. The effect survives correction of the FWER across 5 change point categories.

Previous results obtained from data sampled in the same population using a sparse fNIRS array found evidence for age related differences in the Final AND decreasing (iii) and Latest Decreasing (v) FCPt categories, when applying FCPt detection to cross-infant, group-level haemodynamic responses [10]. The cross-chromophore work described here replicates and extends these findings by (a) using a higher spatial resolution, (b) permitting individual-level analyses, and (c) restricting results to data in which individually-confirmed habituation is present. Consequently, findings here strengthen the previous developmental claim via convergence across methodologies and between cohorts, whilst also accounting for individual variability of habituation between infants [6].

4.4 Chromophore-specific divergence

These cross-chromophore findings appear to have been driven by HbO-specific age differences which were directionally significant ('Final and Decreasing'; iii) and marginally non-significant ('Decreasing', i; 'Earliest Decreasing', iv; and 'Latest Decreasing', v) after correcting the FWER. In contrast, HbR-specific findings showed non-significant but directionally-consistent age effects across all FCPt categories. Group-level TFCE results from previous findings in the same cohort [9] indicated that the prevalence of habituation in infants up to 12mo was not chromophore specific; individual-level TFCE results described in this work seem to support this claim (Table 1 and Figure 4). Thus, it is possible that these findings reflect divergent developmental trajectories in components of neurovascular coupling with differential relevance to each chromophore. For example, maturation of active arteriolar dilation and cerebral blood flow (primarily corresponding to HbO) may proceed on a different timeline to that of cerebral oxygen metabolism (corresponding primarily to HbR) [59], [60].

Alternatively, these findings may reflect a developmental effect on habituation timing that is present across both chromophores but detectable only via HbO in the current experimental setup and at the current sample size [61] – particularly as sample sizes for analyses into HbR-specific FCPts (n = 17-26) were smaller than those for their HbO-specific counterparts (n = 34-44). Future work may seek to investigate the reason for this apparent difference between data for each of the two haemoglobin chromophores, with a clear first step being analysis on a larger number of individual datasets.

4.5 Strengths

Two primary strengths of the work described are rooted in methodological advances. The first of these – the use of HD-DOT – enables individual-level analyses, revealing findings for a subset of the full sample of infants that group averaging obscures [methods paper, zxcv]. This is possible because HD-DOT retains per-participant channel density that supports parcel-space aggregation of individual signals, with the value of this clear from findings obtained using both the TFCE and FCPt detection analyses included in this work.

FCPt detection itself offers further advantages when compared to conventional approaches to fNIRS/HD-DOT analyses. It accounts for correlation between repeated measures and latent temporal structure within repeated evoked haemodynamic responses [62], [63] thus incorporating the trial-specific and temporal information necessary to enable investigation of the gradual change in an evoked haemodynamic response [64]. Ultimately this enables insight into the rate of habituation to repeated stimuli during recording alongside a HaND paradigm as has now been illustrated across separate age-appropriate cohorts, [10], [64], thereby demonstrating its potential utility as an early measure of infant neurodevelopment. This strength is further exemplified when considering age-related differences were uncovered using FCPt detection but not using other approaches. For example, further post-hoc testing for age-related differences in amplitude decrease (Supplementary Materials 5) and haemodynamic response morphology (Supplementary Materials 6) between epochs was also conducted after initial TFCE analyses; these are two features of the changing haemodynamic response which could also be reasonably expected to differ with age [61], [65]. However, no significant age-related differences were found in this sample, suggesting that the developmental effect operates at the level of adaptation timing and is well-captured by FCPt detection.

Further strengths of this work stem from the sample of infants recruited for HD-DOT recording. Firstly, the number of infants recruited is higher than many infant fNIRS/HD-DOT studies [66]; even accounting for the attrition common to infant neuroimaging studies [52] which is also present here, the large sample size of analysed datasets provides a robustness to the findings described. Furthermore, the full sample of infants recruited for the INDiGO Trial will more than double this already large number, meaning future work will further benefit from substantially increased statistical power, potentially enabling detection of effects that did not reach significance in the present sample, such as HbR-based adaptation timing differences. Secondly, catchment area for recruitment – rural Gambia – represents a non-WEIRD (Western, Educated, Industrialised, Rich, Democratic) population that is underrepresented in the developmental cognitive neuroscience literature [24], [67], [68], [69], [70]. Findings obtained from this cohort therefore contribute to the generalisability of habituation as a neurodevelopmental marker beyond the populations in which infant neuroimaging studies are typically conducted.

4.6 Limitations

The low number of ‘responders’ from such a large sample is something which requires further investigation. In addition to the possible effect of the

aforementioned pipeline choices when dealing with infant HD-DOT data, it is also possible that the tile-based design of the particular HD-DOT array used in this study [21], [55], [57] meant that many optodes were prevented from direct contact with the scalp, due to the rigidity of individual tiles preventing simultaneous scalp contact across all optodes on a curved surface, compounded by variation in hair thickness, density, or style. This may have consequently adversely affected the recorded signal [25], thus increasing the possibility of false negatives. Critically, signal-to-noise ratio (SNR) characteristics did not differ meaningfully between ages, ruling out age-dependent noise structure as a driver of the observed timing effect (Supplementary Materials S7). Nevertheless, this remains an open question for the field: whether the low individual-level detection rate reflects a fundamental constraint of infant HD-DOT, or whether it can be substantially improved with optimised hardware and preprocessing.

Furthermore, the paradigm structure itself imposes a constraint on statistical resolution for condition contrast tests. Because each epoch comprises only 5 trials, the label-permutation null for condition contrasts between two 5-trial groups is limited to C(10,5) = 252 possible relabellings, yielding a minimum achievable *p*-value of approximately 0.004. This restricts the sensitivity of individual-level contrast TFCE to detect small effects, potentially leading to false negatives – particularly for infants whose habituation effects are genuine but modest in magnitude, or those with less than the maximum of 5 trials included per condition contrast.

Finally, whilst the ability to obtain individual-level results using the approach outlined in this work is an advance of group-averaged approaches, individual-level results in general are sensitive to sample size and analysis methodology [71]. It is therefore prudent to treat these findings as a step forward that invites verification and extension – including replication across independent cohorts drawn from different populations and, potentially, using varied preprocessing pipelines – before firm conclusions about developmental differences in adaptation timing can be drawn.

5 Conclusion

This work revealed a subset of ‘responder’ and ‘habituator’ infants whose effects were spatially heterogeneous and therefore obscured in group-averaged analyses, reinforcing the idea that group-level null results do not necessarily indicate absence of a developmental effect. Furthermore, it provides individual-level evidence, using HD-DOT, that infant auditory habituation timing differs by age, with 12mo infants habituating significantly earlier than those at 6mo. The convergence of these individual-level findings with earlier group-level, sparse-array results strengthens confidence in habituation timing as a genuine developmental marker, while extending it methodologically to individual-level HD-DOT data. The low number of detected ‘responders’ remains an open question, potentially reflecting methodological/hardware constraints rather than a true absence of neural response, and warrants further investigation. Overall, this study supports HD-DOT, TFCE, and FCPt detection as promising tools for identifying individual-level neurodevelopmental markers in early infancy, with potential future application to early identification developmental trajectories.

Appendices

Appendix A FCPt detection implementation

FCPt detection was conducted using an adaptation of the approach outlined in our previous work using sparse array data [10], which also contains full details on the Monte Carlo simulation, wild binary search implementation, and step-wise holm–Bonferroni multiple comparison correction approaches, which were identical to those conducted in the work reported here.

Functional data was obtained in each parcel location Y for each of the k participants and 25 trials using the *scikit-fda* Python package [72]. Each curve was modelled as:

$$Y_{n,k}(t) = \mu_k(t) + \delta_k(t) \cdot \mathbf{1}_{\{n>n_c\}} + \epsilon_{n,k}(t)$$

Equation A 1

where $1 \leq n \leq 25$ and – for participant k specifically – $\mu_k(t)$ is the mean function, $\epsilon_{n,k}(t)$ is the measurement or systemic error for trial n and $\delta_j(t)$ indicates a structural break at trial $n = n_c$. Candidate change points n_k^* can be identified by $n_k^* = \min\left(arg\ \underset{1\leq n\leq N}{max}\ T_{k,N}^F(n)\right)$ where participant-specific test statistic $T_{k,N}^F$ is given by

$$T_{k,N}^F(n) = \max_{1\leq n\leq N} \frac{1}{N} \int_T \left(P_{k,n}(t)\right)^2 dt = \max_{1\leq n\leq N} \frac{1}{N} \int_T \left(\sum_{i=1}^{n} Y_{i,k}(t) - \frac{n}{N}\sum_{i=1}^{N} Y_{i,k}(t)\right)^2 dt$$

Equation A 2

Here, $Y_{i,k}$ is the functional representation of preprocessed data for the kth participant in the ith trial, in parcel Y. These candidate change points were identified using wild binary segmentation and tested for statistical significance using Monte Carlo simulation. A wild binary search approach was taken within each functional time series $Y_{n,k}$ to account for, and identify, possible multiple FCPts [73]. During wild binary segmentation, if a statistically significant FCPt is detected on an interval (segment) of data for a particular $Y_{n,k}$, the interval is split into two subintervals. For each of these two subintervals, multiple pseudo-subintervals of different lengths are generated and FCPt detection is conducted on each to detect pseudo-candidate FCPts and their associated pseudo-test statistics. The candidate change point n^* and its associated $T_{k,N}^F(n^*)$ for the entire data subinterval corresponds to the location of the largest test statistic across all pseudo-subintervals. The process repeats recursively until no significant FCPts remain.

During Monte Carlo simulation of the null distribution, $M = 10\,000$ summed Brownian Bridges. The maximum value for the mth Brownian Bridge, for participant k's data, in parcel Y, $B_{k,m}$ was given by

$$B_{k,m} = \max_{0\leq n\leq N_b} \sum_{\xi=1}^{d} \lambda_\xi B_\xi^2(n)$$

Equation A 3

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where $B_\xi(n)$ are Brownian Bridges the same length in time as the recorded data for parcel Y; λ_ξ are their associated sample eigenvalues under spectral decomposition of the residuals $\epsilon_{n,k}(t)$ under the estimated covariance operator $c(t,s) = \frac{1}{N}\sum_{1\le n\le N}\{Y_{n,k}(t) - \bar{Y}_{N,k}(t)\}\{Y_{n,k}(s) - \bar{Y}_{N,k}(s)\}$, where $\bar{Y}_{N,k}(t)$ is the sample mean function; and N_b is the length of the interval being assessed at each stage of the recursive wild binary segmentation process. The statistical significance of $T^F_{k,N}(n^*)$ – the identified FCPt in parcel Y for participant k – was then given by the proportion of the M randomly generated values larger than $T^F_{k,N}(n^*)$.

Given the hierarchical nature of the wild binary segmentation process, in which the importance of change points at higher levels is predicated on the statistical significance beforehand, a step-down multiple-comparison procedure was conducted within each channel at every wild binary search level to correct the family-wise error rate (FWER), using the Holm–Bonferroni method ($\alpha = 0.01$).

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Declaration of competing interests

The authors have no conflict of interest to declare.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work the author(s) used Claude in order to scan for typos and ensure consistent use of British English. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Code Availability

The code used to conduct the analyses and generate the figures presented in this paper are available at https://github.com/sam-beaton/fcpt and rely on the following open source Python packages: *h5py* [74], *numpy* [75] , *matplotlib* [76], *nibabel* [77], *os* [78], *pandas* [79], *pickle* [80], *plotly* [81], *scikit-fda* [72], *scipy* [48], *seaborn* [82], *statistics* [83], *statsmodels* [46], *sys* [84].

Data Availability

The data used to support this study are stored in…(to fill in later) zxcv

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Biographies

Samuel Beaton is a researcher in the Department of Women and Children's Health at King's College London. His research involves developing analytical pipelines for functional near-infrared spectroscopy (fNIRS) and high-density diffuse optical tomography (HD-DOT) data, using these to investigate brain development in infancy.

Samantha McCann is a Public Health Registrar and formerly Postdoctoral Research Associate in the Department of Women and Children's Health at King's College London. Her main research interest is supporting early child development, with a strong focus on the impact of undernutrition in infancy on long-term neurodevelopmental outcomes.

Ebrima Mbye is a Field Coordinator at MRCG@LSHTM currently engaged in the INDiGO Trial working under Professor Sophie Moore.

Anna Blasi is a Postdoctoral Research Fellow at UCL. Her research interests are centred on functional aspects of human physiology. Her research career started with models of the cardiovascular system and the effects of disease. Through her work at UCL, KCL, and Birkbeck, her research interests have shifted toward the use of functional imaging (fNIRS, fMRI) to study brain function and neurocognitive development in early infancy.

Sophie Moore is Professor of Global Women and Children's Health in the Department of Women & Children's Health at King's College London and an Honorary Associate Professor at the London School of Hygiene and Tropical Medicine (LSHTM). Her research focuses on the nutritional regulation of 'healthy' fetal and infant growth, incorporating infant immune and brain development outcomes, and on the mechanisms through which maternal, infant and childhood nutrition may influence development and later health.

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Tables

Table 1: Placeholder

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Placeholder

Figure Captions

Figure 1: Ignore for now – placeholder for later on.

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Supplementary Materials

S1. Cross-infant, group-averaged response results

The following results for 6mo and 12mo infants were obtained using two-sample *t*-tests on data from the same 250 infants. Full methodology and results can be found in the publication [methods paper, zxcv] but a summary of the approach and findings are given here for clarity and ease of access.

Methods

Parcel-space data were analysed at the group-level on cross-infant, group-averaged signals for each age, with analysis conducted on both HbO and HbR signals. Paired t-tests were performed to identify the presence and spatial distribution of significant haemodynamic responses, as well as their modulation across the experimental paradigm, following approaches used in previous studies [16].

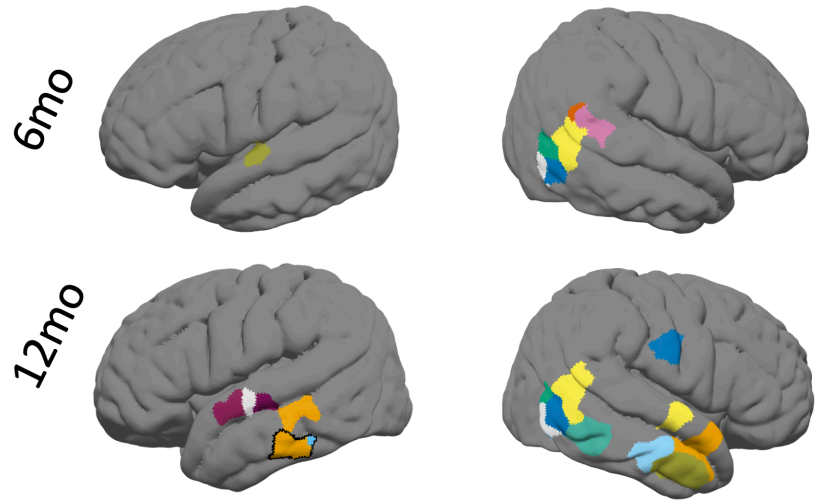
For group-level analysis, a single response value per participant per condition was calculated by averaging the signal within the response window (8 – 12s post-stimulus onset) across all valid trials for that condition. Baseline values were similarly computed per participant by averaging the pre-stimulus window (-4 – 0s) across all Fam1 trials. A parcel × chromophore combination at a given age was included in testing only if data from at least 5 participants were available, with each participant required to contribute at least 3 valid trials within the relevant epoch.

Significant haemodynamic responses were assessed using one-tailed paired t-tests comparing baseline and Fam1 response values, testing for a positive peak response in HbO and a negative peak response in HbR. Parcel × chromophore combinations showing significant activation were then tested for the same condition contrasts and using the same prerequisite chain thresholding as specified in the main text. All t-test results were corrected for multiple comparisons following each testing stage using the Holm–Bonferroni procedure [51], with a family-wise error rate of $\alpha = 0.05$, implemented using the statsmodels Python package [46].

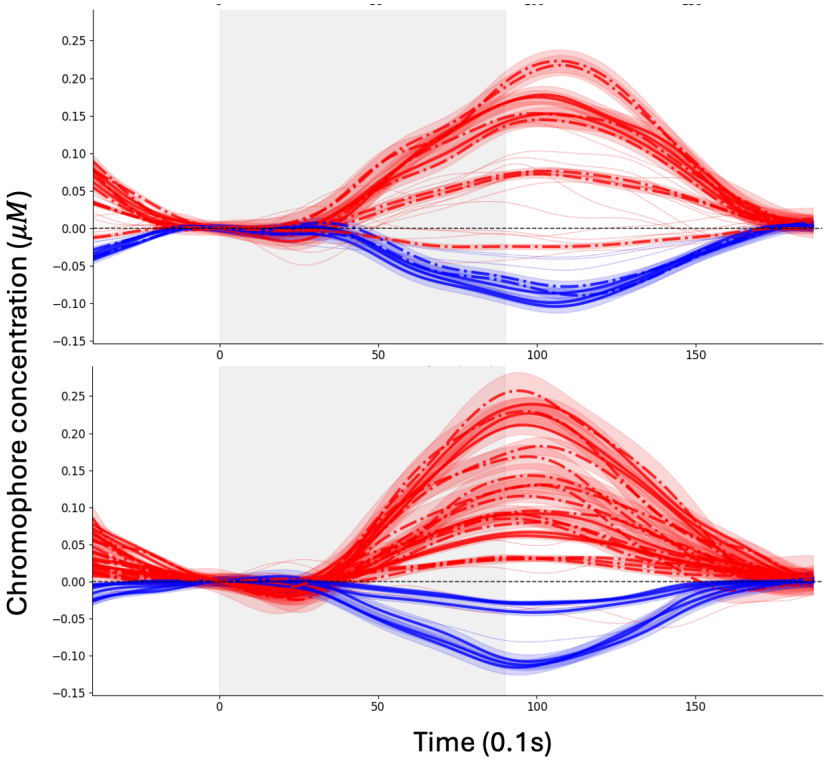
Results

Significant task-evoked haemodynamic responses were identified in 16 and 43 parcel–chromophore combinations at 6- and 12mo, respectively. Significant responses were largely localised to temporal regions and the superior temporal sulcus, consistent with the auditory-associative nature of the HaND paradigm.

Significant habituation) was observed in 2 parcel × chromophore combinations at 12mo, both in HbO and both localised to right-hemisphere temporal regions(Figures S1 and S2). No significant habituation was detected at 6mo. Delayed habituation was observed in the same two parcels that showed significant habituation at 12mo, suggesting that the response remained suppressed relative to Fam1 throughout the novelty epoch. No significant novelty detection response was identified at either age.



Caption: Location in the cerebral cortex of parcels containing significant evoked haemodynamic responses in either chromophore at each age. The two outlined parcels in the left hemisphere at 12mo exhibited delayed activation.



Caption: chromophore concentration change in parcels with significant evoked responses. Stimulus onset at time = 0s; red = HbO; blue = HbR. Solid lines represent parcels with significant responses in both chromophores. Dash-dot lines represent parcels where only one of the two chromophores exhibit a significant evoked response. Faint dashed lines represent concentration changes in parcels which exhibited significant responses in at least one chromophore for at least one other timepoint.

S2. Cross-level concordance between cross-infant, group-averaged results and individual-level TFCE results

Methods

To examine whether individual-level metrics were associated with group-level significance at the parcel level, two complementary analyses were performed.

For each parcel, individual-level summary metrics were computed from the per-infant TFCE results: the number of individually-significant infants, the proportion of tested infants reaching individual-level significance, median individual *t*-statistic, and median individual TFCE peak value. Group-level significance was defined as surviving FDR correction ($q < 0.05$) in the cross-participant paired t-test (Methods and Supplementary Materials S1).

First, Mann–Whitney U tests compared each individual-level metric between parcels that reached group-level significance and those that did not (one-sided, H_1: group-significant parcels have higher individual-level metrics). *P*-values were Holm–Bonferroni-corrected across the four metrics within each age × chromophore combination. Second, logistic regression was fitted with group-level significance (binary) as the outcome and the four individual-level summary metrics as predictors. Variance inflation factors (VIF) were computed prior to model fitting to assess multicollinearity among predictors; any predictor with VIF > 5 was flagged and, if necessary, removed. Model coefficients were exponentiated to obtain odds ratios with 95% confidence intervals. Predictor p-values were Holm–Bonferroni-corrected across the four coefficients within each age × chromophore model.

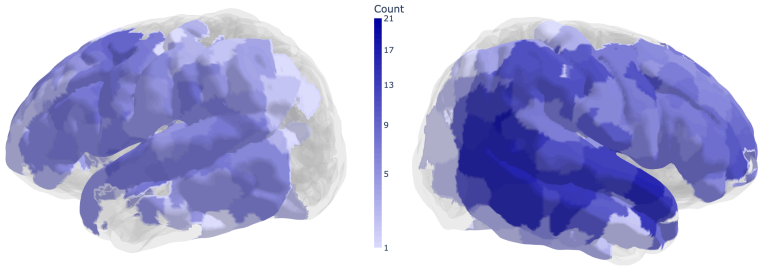
Results

Mann–Whitney U tests comparing individual-level metrics between group-significant and group-non-significant parcels produced highly consistent results across all four age × chromophore combinations. Group-significant parcels had substantially more individually-significant infants (median 13–17 vs 4–6 at group-non-significant parcels), a higher proportion of individually-significant infants, higher median individual t-statistics, and higher median individual TFCE peaks (all *p(corrected)* < 0.001 for all individual metrics). Critically, the number of infants tested at each parcel did not differ between group-significant and group-non-significant parcels at three of four age × chromophore combinations (6mo HbO: *p(corrected)* = 0.056; 6mo HbR: *p(corrected)* = 0.135; 12mo HbR: *p(corrected)* = 0.320), with a small but significant difference observed only at 12mo HbO (median 175 vs 167 *p(corrected)* = 0.005).

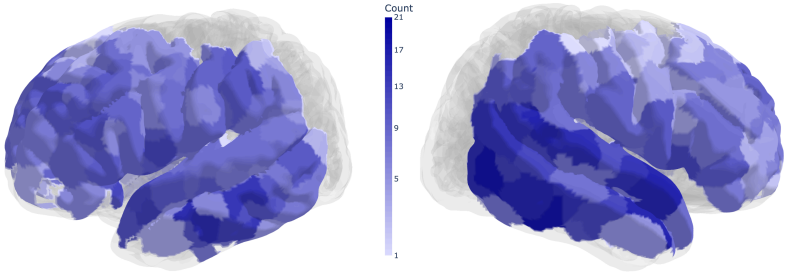
Logistic regression identified the number of individually-significant infants and the median individual TFCE peak as significant predictors of group-level significance for HbO at both 6mo (n_infants_sig: β = +0.84, *p(corrected)* = 0.004; median TFCE peak: β = +0.004, *p(corrected)* = 0.022) and 12mo (n_infants_sig: β = +0.42, *p(corrected)* < 0.001; median TFCE peak: β = +0.002, *p(corrected)* = 0.006). Neither the proportion of infants individually significant nor the median individual *t*-statistic survived correction at either age. For HbR, all predictors showed directionally consistent effects but none survived correction, likely reflecting the smaller number of group-significant parcels available for HbR (7–8 vs 11–25 for HbO).

S3. HbR significant responses and habituation at the individual level

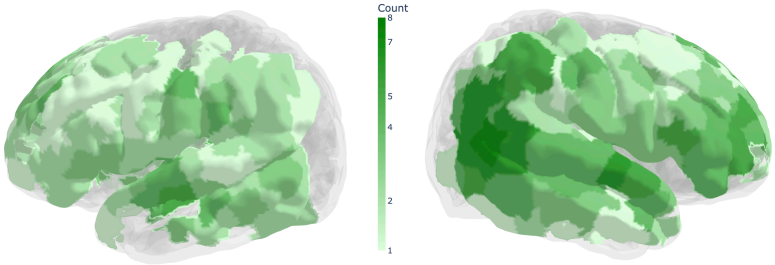
6mo – significant haemodynamic responses



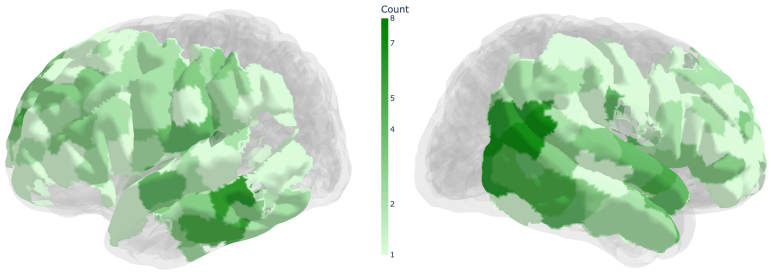
12mo – significant haemodynamic responses



6mo – significant habituation



12mo – significant habituation



S4. Age differences in FCPt timing – wider group of 'responders'

		Ordinal logistic regression			Wald test	Tukey HSD (12 vs 6mo)	
	FCPt category	**β**	**p (raw)**	**p (Holm)**	**p**	**Δ trials**	**p**
Cross-chromophore	Decreasing	-0.1140	0.1093	0.3280	0.1093	-1.5361	0.1032
	First and decreasing	-0.1186	0.1571	0.3280	0.1571	-1.5810	0.1090
	Final and decreasing	**-0.2242**	**0.0068****	**0.0341***	**0.0068****	**-2.9481**	**0.0034****
	Earliest decreasing	-0.1060	0.1625	0.3280	0.1625	-1.1982	0.2064
	Latest decreasing	-0.1494	0.0517	0.2069	0.0517	-2.0671	0.0348
HbO	Decreasing	-0.2309	**0.0184***	0.0735	0.0184	-2.8125	0.0129
	First and decreasing	-0.2088	0.0549	0.0867	0.0549	-2.2274	0.0408
	Final and decreasing	**-0.3207**	**0.0051****	**0.0256***	**0.0051****	**-3.8632**	**0.0017****
	Earliest decreasing	-0.2255	**0.0289***	0.0867	0.0289	-2.2417	0.0256
	Latest decreasing	-0.2223	**0.0302***	0.0867	0.0302	-2.8500	0.0196
HbR	Decreasing	-0.0124	0.9097	1	0.9097	-0.0240	0.9882
	First and decreasing	-0.0220	0.8710	1	0.8710	-0.6250	0.7484
	Final and decreasing	-0.1627	0.2067	1	0.2067	-2.0606	0.2034
	Earliest decreasing	-0.0233	0.8433	1	0.8433	-0.1859	0.9163
	Latest decreasing	-0.1004	0.4061	1	0.4061	-1.4872	0.3299

*OLR with Age as fixed effect applied to each FCPt category. OLR p-values Holm–Bonferroni corrected across 5 tests within each chromophore group. Wald and Tukey HSD reported uncorrected as follow-ups. β = OLR coefficient for Age; negative values indicate earlier change point trial numbers at 12 months compared to 6 months. Δ trials = Tukey HSD mean difference (12mo − 6mo); negative values give approximate trial number differences between ages for each FCPt category. * signifies p < 0.05; ** signifies p< 0.01.*

S5. Age-related differences in habituation amplitude

S5.1 Methods

To test for age differences in habituation magnitude, the difference between the mean Fam1 peak-window (8–12s) amplitude and the mean contrast-group peak-window amplitude was calculated to give the habituation amplitude for each infant's personal response parcel set waveform. For HbR, this quantity was sign-inverted so that positive values reflect canonical habituation for both chromophores. Wilcoxon signed-rank tests assessed whether habituation amplitude was significantly different from zero at each age × chromophore (one-sided: greater than zero for HbO; less than zero for HbR, prior to sign-inversion). Mann–Whitney U tests compared habituation amplitude between 6- and 12mo infants within each chromophore. Analyses were performed for three contrasts: Fam1 vs Fam3 (standard habituation), Fam1 vs Novel (delayed habituation), and Fam1 vs Post-test (long-range habituation).

S5.2 Results

For all three condition contrasts, population-level habituation was robust for both chromophores at both ages:

Contrast	Age	Chromophore	n	W	*p*	Median (μM)
Habituation	6mo	HbO	29	400	<0.001	0.172
		HbR	25	6	<0.001	-0.129
	12mo	HbO	35	565	<0.001	0.180
		HbR	34	4	<0.001	-1.20
Delayed habituation	6mo	HbO	30	451	<0.001	0.197
		HbR	26	10	<0.001	-0.139
	12mo	HbO	36	622	<0.001	0.234
		HbR	36	31	<0.001	-0.103
Long-range habituation	6mo	HbO	29	406	<0.001	0.239
		HbR	26	20	<0.001	-0.119
	12mo	HbO	33	523	<0.001	0.149
		HbR	34	36	<0.001	-0.066

Mann-Whitney U tests showed no significant age difference for any condition contrast × chromophore combination:

Condition contrast	Chromophore	U	*p*
Habituation	HbO	514	0.47
	HbR	430	0.47
Delayed habituation	HbO	546	0.47
	HbR	396	0.81
Long-range habituation	HbO	581	0.075
	HbR	336	0.94

S6. Age-related differences in haemodynamic response morphology

Methods

To test for the possibility of age-related differences in haemodynamic response shape or latency under habituation, two supplementary analyses were performed on the individual response parcel set waveforms.

(1) Waveform correlation.

For each infant, Pearson correlation coefficients were computed between the average individual response parcel set waveform for Fam1 and contrast-groups over three windows: a wide window (0-15 s post-stimulus), a tight window (5-13 s post-stimulus), and a lag-optimised comparison, where the contrast-group waveform was shifted by lags of ±3 s (in 0.1 s steps) and the maximum correlation retained. The latter correlation quantifies the best achievable shape agreement between the two waveforms, regardless of any timing offset. Between-age comparisons used Mann–Whitney U tests (one-sided: H_1 = 6mo > 12mo).

(2) Peak Latency.

For each infant, the peak time of the average individual response parcel set Fam1 waveform and of the contrast-group waveform was identified within a 0-13s post-stimulus window. The difference (contrast-group peak time minus Fam1 peak time) was computed per infant. Wilcoxon signed-rank tests assessed whether this difference was significantly greater than zero (i.e. whether the contrast-group response peaked later than Fam1) at each age.

Results

(1) Waveform correlation

Lag-optimised Fam1 vs Fam3 waveform correlations were high and comparable between ages for both chromophores (HbO: 6mo median r = 0.786, 12mo median r = 0.815, Mann–Whitney p = 0.83; HbR: 6mo median r = 0.748, 12mo median r = 0.719, p = 0.30). Similar patterns were observed for Fam1 vs Novel (HbO: 6mo r = 0.791, 12mo r = 0.812, p = 0.66; HbR: 6mo r = 0.840, 12mo r = 0.750, p = 0.22) and Fam1 vs Post-test (HbO: 6mo r = 0.704, 12mo r = 0.828, p = 0.89; HbR: 6mo r = 0.732, 12mo r = 0.688, p = 0.60). No comparisons reached significance, indicating comparable waveform shapes across ages.

Wide-window and tight-window (zero-lag) correlations were substantially lower (medians <~0.3), reflecting the noise-diluting effect of including non-response time points, but also showed no age differences.

(2) Peak latency

No systematic age-dependent latency shift was observed between:

- Fam1 vs Fam3 (HbO: 6mo median = −1.50 s, Wilcoxon p = 0.97; 12mo median = 0.90 s, p = 0.07; HbR: 6mo median = 0.00 s, p = 0.46; 12mo median = −0.75 s, p = 0.37);

- Fam1 vs Novel (HbO: 6mo median = −3.50 s, $p > 0.99$; 12mo median = −1.80 s, p = 0.97; HbR: 6mo median = −2.20 s, $p = 0.96$; 12mo median = 0.90 s, p = 0.30); or
- Fam1 vs Post-test (HbO: 6mo median = 0.30 s, p = 0.84; 12mo median = −0.10 s, $p = 0.62$; HbR: 6mo median = −0.45 s, p = 0.34; 12mo median = 1.85 s, $p = 0.13$).

S7. Age-dependent signal quality

S7.1 Methods

The SNR of each averaged individual response parcel set waveform was calculated for each epoch and chromophore. SNR was defined as the expected peak amplitude in the 8–12 s post-stimulus window divided by the standard deviation of the pre-stimulus baseline (-40–0s), and was computed separately for the Fam1 waveform and the contrast-group waveforms (Fam3, Novel, or Post-test) for each infant × chromophore combination. Between-age comparisons were performed using two-sided Mann–Whitney U tests within each contrast × chromophore × epoch combination.

S7.2 Results

No between-age SNR comparison reached significance at any contrast, chromophore, or epoch (all p > 0.30, two-sided Mann–Whitney U):

- Fam1 v Fam3: For HbO, median Fam1 SNR was 7.84 at 6mo (n = 29) and 6.61 at 12mo (n = 35; Mann–Whitney U = 542, p = 0.65); median Fam3 SNR was 5.48 at 6mo and 4.51 at 12mo (U = 580, p = 0.33). For HbR, median Fam1 SNR was 6.75 at 6mo (n = 25) and 6.34 at 12mo (n = 34; U = 493, p = 0.30); median Fam3 SNR was 3.56 at 6mo and 4.65 at 12mo (U = 377, p = 0.47).
- Fam1 v Novel: For HbO, median Fam1 SNR was 7.86 at 6mo (n = 30) and 6.35 at 12mo (n = 36; U = 589, p = 0.53); median Novel SNR was 4.30 at 6mo and 4.96 at 12mo (U = 517, p = 0.77). For HbR, median Fam1 SNR was 6.41 at 6mo (n = 26) and 6.37 at 12mo (n = 35; U = 489, p = 0.63); median Novel SNR was 4.30 at 6mo and 3.33 at 12mo (U = 497, p = 0.55).
- Fam1 v Post-test: For HbO, median Fam1 SNR was 7.84 at 6mo (n = 29) and 6.09 at 12mo (n = 33; U = 531, p = 0.46); median Post-test SNR was 3.49 at 6mo and 4.18 at 12mo (U = 410, p = 0.34). For HbR, median Fam1 SNR was 6.41 at 6mo (n = 26) and 6.34 at 12mo (n = 34; U = 478, p = 0.60); median Post-test SNR was 4.82 at 6mo and 4.13 at 12mo (U = 508, p = 0.33).